

PROGRAM 1

AIM: Write program to check whether the number is in the range.

SOURCE CODE

```
number = int(input("Enter a number: "))
lower_bound = int(input("Enter the lower bound: "))
upper_bound = int(input("Enter the upper bound: "))
if lower_bound <= number <= upper_bound:
    print(f"The number {number} is within the range {lower_bound} to {upper_bound}.")
else:
    print(f"The number {number} is NOT within the range {lower_bound} to {upper_bound}.")
```

OUTPUT

```
===== RESTART: C:\Users\devik\Downloads\devika\range.py ==  
Enter a number: 5  
Enter the lower bound: 2  
Enter the upper bound: 10  
The number 5 is within the range 2 to 10.  
  
===== RESTART: C:\Users\devik\Downloads\devika\range.py ==  
Enter a number: 2  
Enter the lower bound: 10  
Enter the upper bound: 20  
The number 2 is NOT within the range 10 to 20.  
|
```

PROGRAM 2

AIM: Calculate the distance between two points taking input from the user and using Pythagoras theorem.

SOURCE CODE

```
import math

x1 = float(input("Enter the x1: "))
y1 = float(input("Enter the y1: "))
x2 = float(input("Enter the x2: "))
y2 = float(input("Enter the y2: "))

distance = math.sqrt((x2 - x1)**2 + (y2 - y1)**2)

print(f"The distance between the points ({x1}, {y1}) and ({x2}, {y2}) is {distance:.2f}")
```

OUTPUT

```
===== RESTART: C:\Users\devik\Downloads\devika\distance.py ==  
Enter the x1: 5  
Enter the y1: 10  
Enter the x2: 15  
Enter the y2: 20  
The distance between the points (5.0, 10.0) and (15.0, 20.0) is 14.14  
|
```

PROGRAM 3

AIM: Accept the no of days, the member is late to return the book and display the fine or appropriate message.

SOURCE CODE

```
days_late = int(input("Enter the number of days the book is returned  
late: "))  
  
if days_late <= 3:  
    print("No fine. Thank you for returning the book on time!")  
  
elif days_late <= 10:  
    print("Fine: ₹1 per day")  
    print(f"Total fine: ₹{days_late * 1}")  
  
elif days_late <= 20:  
    print("Fine: ₹2 per day")  
    print(f"Total fine: ₹{days_late * 2}")  
  
elif days_late <= 30:  
    print("Fine: ₹5 per day")  
    print(f"Total fine: ₹{days_late * 5}")  
  
else:  
    print("Membership is cancelled due to delay of more than 30  
days.")
```

OUTPUT

```
===== RESTART: C:\Users\devik\Downloads\devika\library.py =====  
Enter the number of days the book is returned late: 2  
No fine. Thank you for returning the book on time!  
  
===== RESTART: C:\Users\devik\Downloads\devika\library.py =====  
Enter the number of days the book is returned late: 10  
Fine: ₹1 per day  
Total fine: ₹10  
  
===== RESTART: C:\Users\devik\Downloads\devika\library.py =====  
Enter the number of days the book is returned late: 31  
Membership is cancelled due to delay of more than 30 days.  
|
```

PROGRAM 4

AIM: Print n terms of Fibanocci series of numbers.

SOURCE CODE

```
limit = int(input("Enter the limit for Fibonacci numbers: "))  
a, b = 0, 1  
print("Fibonacci numbers up to", limit, ":")  
while a <= limit:  
    print(a, end=" ")  
    a, b = b, a + b
```

OUTPUT

```
===== RESTART: C:/Users/devik/Downloads/devika/fibonacci.py =====
```

```
Enter the limit for Fibonacci numbers: 6
```

```
Fibonacci numbers up to 6 :
```

```
0 1 1 2 3 5
```

```
|
```


PROGRAM 5

AIM: Find all prime numbers in given range.

SOURCE CODE

```
start = int(input("Enter start range: "))
end = int(input("Enter end range: "))
print(f"Prime numbers between {start} and {end} are:")
for num in range(start, end + 1):
    if num > 1:
        for i in range(2, num):
            if (num % i) == 0:
                break
        else:
            print(num)
```

OUTPUT

```
===== RESTART: C:/Users/devik/Downloads/devika/prime.py =====  
Enter start range: 4  
Enter end range: 20  
Prime numbers between 4 and 20 are:  
5  
7  
11  
13  
17  
19
```

PROGRAM 6

AIM: Print date and time for today and now.

SOURCE CODE

```
from datetime import datetime  
now = datetime.now()  
print("Today's Date:", now.date())  
print("Current Time:", now.time())
```

OUTPUT

```
===== RESTART: C:/Users/devik/Downloads/devika/dateandtime.py =====  
Today's Date: 2025-07-26  
Current Time: 11:06:33.628315
```

PROGRAM 7

AIM: Create a list and perform following methods.

insert(), append(), len(), pop(), clear()

SOURCE CODE

```
arr = list(map(int, input("Enter space seperated numbers: ").split()))
print("Created new list:", arr)
pos = int(input("Enter position to insert: "))
num = int(input("Enter number: "))
arr.insert(pos, num)
print(f"Inserted {num} at index position {pos}:", arr)
num = int(input("Enter a number to remove: "))
arr.remove(num)
print(f"Removed the first occurence of {num} from list:", arr)
num = int(input("Enter a number to append: "))
arr.append(num)
print(f"Appended {num} to the list:", arr)
print("Length of list =", len(arr))
arr.pop()
print("Poped the last element of list:", arr)
arr.clear()
print("Cleared the list:", arr)
```

OUTPUT

```
===== RESTART: C:/Users/devik/Downloads/devika/list.py =====  
Enter space seperated numbers: 1 2 3 4 5  
Created new list: [1, 2, 3, 4, 5]  
Enter position to insert: 2  
Enter number: 7  
Inserted 7 at index position 2: [1, 2, 7, 3, 4, 5]  
Enter a number to remove: 3  
Removed the first occurence of 3 from list: [1, 2, 7, 4, 5]  
Enter a number to append: 9  
Appended 9 to the list: [1, 2, 7, 4, 5, 9]  
Length of list = 6  
Poped the last element of list: [1, 2, 7, 4, 5]  
Cleared the list: []  
>
```

PROGRAM 8

AIM: Create a menu with following options

Perform addition, subtraction, multiplication, division, mod division, integer division.

SOURCE CODE

while True:

```
    print("\n--- Simple Calculator Menu ---")
```

```
    print("1. Addition")
```

```
    print("2. Subtraction")
```

```
    print("3. Multiplication")
```

```
    print("4. Division")
```

```
    print("5. Modulus (a % b)")
```

```
    print("6. Integer Division (a // b)")
```

```
    print("7. Exit")
```

```
    choice = input("Enter your choice (1-7): ")
```

```
    if choice in ['1', '2', '3', '4', '5', '6']:
```

```
        try:
```

```
            a = float(input("Enter first number: "))
```

```
            b = float(input("Enter second number: "))
```

```
        except ValueError:
```

```
            print("Invalid input. Please enter numbers.")
```

```
            continue
```

```
if choice == '1':
    print(f"Result: {a} + {b} = {a + b}")
elif choice == '2':
    print(f"Result: {a} - {b} = {a - b}")
elif choice == '3':
    print(f"Result: {a} * {b} = {a * b}")
elif choice == '4':
    if b != 0:
        print(f"Result: {a} / {b} = {a / b}")
    else:
        print("Error: Division by zero.")
elif choice == '5':
    if b != 0:
        print(f"Result: {a} % {b} = {a % b}")
    else:
        print("Error: Modulus by zero.")
elif choice == '6':
    if b != 0:
        print(f"Result: {a} // {b} = {a // b}")
    else:
        print("Error: Integer division by zero.")
elif choice == '7':
    print("Exiting calculator. Goodbye!")
    break
```


else:

```
print("Invalid choice. Please enter a number from 1 to 7.")
```

OUTPUT

```
=====
--- Simple Calculator Menu ---
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Modulus (a % b)
6. Integer Division (a // b)
7. Exit
Enter your choice (1-7): 1
Enter first number: 20
Enter second number: 30
Result: 20.0 + 30.0 = 50.0

--- Simple Calculator Menu ---
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Modulus (a % b)
6. Integer Division (a // b)
7. Exit
Enter your choice (1-7): 2
Enter first number: 20
Enter second number: 10
Result: 20.0 - 10.0 = 10.0

--- Simple Calculator Menu ---
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Modulus (a % b)
6. Integer Division (a // b)
7. Exit
Enter your choice (1-7): 3
Enter first number: 20
Enter second number: 30
Result: 20.0 * 30.0 = 600.0

--- Simple Calculator Menu ---
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Modulus (a % b)
6. Integer Division (a // b)
7. Exit
Enter your choice (1-7): 4
Enter first number: 400
Enter second number: 4
Result: 400.0 / 4.0 = 100.0

--- Simple Calculator Menu ---
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Modulus (a % b)
6. Integer Division (a // b)
7. Exit
Enter your choice (1-7): 5
Enter first number: 70
Enter second number: 4
Result: 70.0 % 4.0 = 2.0
```

OUTPUT

```
--- Simple Calculator Menu ---
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Modulus (a % b)
6. Integer Division (a // b)
7. Exit
Enter your choice (1-7): 6
Enter first number: 789
Enter second number: 3
Result: 789.0 // 3.0 = 263.0

--- Simple Calculator Menu ---
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Modulus (a % b)
6. Integer Division (a // b)
7. Exit
Enter your choice (1-7): 7
Exiting calculator. Goodbye!
|
```

PROGRAM 9

AIM: Find factorial using function.

SOURCE CODE

```
def factorial(n):  
    if n == 0 or n == 1:  
        return 1  
  
    f = 1  
    for i in range(2, n + 1):  
        f *= i  
  
    return f  
  
num = int(input("Enter a number: "))  
print("Factorial is:", factorial(num))
```

OUTPUT

```
=====
Enter a number: 5
Factorial is: 120

```